

SIMATS ENGINEERING
ASSESSMENT TOOL 3: MCQ Test

COURSE NAME: COMPUTER VISION

COURSE CODE: DSA0204

COURSE OUTCOME COVERED

CO3: *Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BTL 4)*

Assessment Tool Weightage: 10%

Code of Conduct:

I K Sampath Reddy (192424285) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action

Signature of the student: K Sampath Reddy

RUBRICS FOR CALCULATION:

			mistakes	errors	
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions

No .	Question	A	B	C	D
1	Which field primarily deals with extracting meaningful information from digital images and videos?	Computer Vision	Database Management	Operating Systems	Computer Networks
2	Image processing mainly focuses on:	Understanding human emotions	Manipulating and enhancing images	Designing computer hardware	Managing databases
3	Computer graphics is primarily concerned with:	Creating and rendering visual content	Detecting diseases	Recognizing handwriting	Tracking vehicles
4	Which level of computer vision deals mainly with basic image features such as edges and intensity?	High-level vision	Mid-level vision	Low-level vision	Semantic vision
5	Which is an example of highlevel computer vision?	Image smoothing	Edge detection	Object recognition	Noise removal
6	Biometrics commonly uses computer vision for:	Identifying individuals	Compressing files	Formatting images	Rendering animations
7	Which application retrieves images based on their visual content?	CBIR	OCR	GPS	CAD

8	The pinhole camera model assumes that:	Light rays are randomly scattered	Light travels in straight lines through a small aperture	Images are always 3D	Objects emit light uniformly
9	Camera calibration is primarily used to determine:	Image file size	Camera parameters	Image compression ratio	Display resolution
10	Which is an intrinsic camera parameter?	Focal length	Object velocity	Object depth	World coordinate
11	Binocular imaging systems generally use:	One camera	Two cameras	Four cameras	No cameras
12	Stereo vision estimates depth primarily using:	Color information	Differences between corresponding image points	Image brightness alone	Image compression
13	Multiple View Geometry studies relationships between:	Multiple images and camera viewpoints	Audio signals	Database tables	Computer processors
14	Structure from Motion	Image color only	3D structure and	Image	Audio

	attempts to recover:		camera motion	compression parameters	frequencies
15	Shape from shading estimates surface shape using variations in:	Color names	Image intensity	File format	Pixel coordinates only
16	Photometric stereo is primarily used to estimate:	Surface orientation	Camera resolution	Image file size	Object category
17	Depth from Defocus determines depth using:	Image blur information	Object color	Image histogram only	Camera brand
18	Which technology combines real-world scenes with computer-generated information?	Virtual Memory	Augmented Reality	Database Systems	Operating Systems
19	A 3D model constructed from images represents:	Only image brightness	Three-dimensional geometric information	Only pixel locations	Only camera settings
20	Medical image analysis is commonly applied to:	Disease diagnosis and anatomical analysis	Video compression only	Network routing	Software compilation
No .	Question	A	B	C	D
21	Image preprocessing is generally performed to:	Improve image quality before further analysis	Delete all image information	Convert images into text automatically	Increase camera focal length
22	Which operation is commonly used for reducing image noise?	Filtering	Object recognition	Camera calibration	Feature matching
23	A continuous image can be represented mathematically as:	A function of continuous spatial coordinates	A fixed-length text string	A database table	A binary program
24	A digital image is generally represented as:	A discrete array of pixels	A continuous sound wave	A mathematical equation only	A network graph only
25	Which technique is commonly used to detect abrupt intensity changes?	Edge detection	Image compression	Color quantization	Image resizing

26	The Sobel operator is primarily used for:	Edge detection	Object classification	Image generation	Image compression
27	The Canny edge detector is designed to:	Detect edges effectively while reducing noise sensitivity	Generate 3D models	Classify objects using neural networks	Perform database searches
28	An image feature should ideally be:	Useful for distinguishing objects or regions	Completely random	Independent of image content	Always represented by text
29	Which is a local image feature?	Corner	Image file name	Camera serial number	File extension
30	Feature extraction is performed to:	Obtain meaningful characteristics from images	Increase image noise	Remove all image information	Change computer hardware
31	Optical flow represents:	Apparent motion of image brightness patterns	Image color distribution	Camera resolution	Object classification labels
32	Optical flow is particularly useful for estimating:	Motion between image frames	Image file size	Camera aperture only	Object category
33	Regularization in motion estimation is used mainly to:	Stabilize an ill-posed estimation problem	Increase image brightness	Change image dimensions	Remove camera parameters
34	Stereo vision uses multiple images to estimate:	Depth	File size	Image format	Pixel color names
35	Correspondence in stereo vision refers to:	Finding matching points in different images	Changing image resolution	Removing noise	Generating random pixels
36	Motion estimation attempts to	Movement of objects	Image	Pixel storage	Camera

	determine:	or image regions	compression ratio	requirements	manufacturing details
37	Structure from Motion uses image sequences to recover:	Camera motion and scene structure	Image file names	Pixel colors only	Audio information
38	Which feature detector is particularly associated with scale and rotation invariance?	SIFT	JPEG	MPEG	PNG
39	A feature descriptor is used to:	Represent the characteristics of an image feature	Store the image file extension	Control camera exposure	Increase network bandwidth
40	Which preprocessing operation changes the spatial dimensions of an image?	Resizing	Histogram equalization	Thresholding	Edge detection
No .	Question	A	B	C	D
41	Shape representation describes:	Geometric characteristics of an object	File compression methods	Camera battery level	Image file size only
42	A contour-based representation primarily uses:	Object boundaries	Object color only	Background intensity	Camera parameters
43	Region-based shape representation considers:	Pixels contained within an object region	Only the object's boundary	Camera motion	Image metadata
44	Which representation focuses mainly on the boundary of an object?	Contour representation	Region representation	Histogram representation	Texture representation

45	Deformable models are useful for representing:	Objects whose shapes can change	Only rigid objects	Image file formats	Camera calibration matrices
46	Snakes are also known as:	Active contours	Static contours	Binary filters	Feature histograms
47	Active contours evolve under the influence of:	Internal and external forces	File compression parameters	Camera serial numbers	Network protocols
48	The main purpose of snakes is to:	Locate object boundaries	Compress images	Generate random images	Increase image resolution
49	Level set methods represent contours using:	An implicit function	A text file	A color palette	A camera lens
50	One advantage of level set representations is their ability to handle:	Topological changes	File conversion	Camera manufacturing	Image printing
51	Fourier descriptors are commonly used for:	Shape representation	Image encryption	Camera calibration	Audio processing
52	Fourier descriptors represent shape using:	Frequency-domain coefficients	Pixel file names	Camera parameters	RGB values only
53	Wavelet descriptors provide information in:	Multiple scales and frequencies	Only one spatial location	Camera coordinates only	File metadata
54	Medial representation is associated with the:	Skeleton or medial axis of an object	Object color palette	Camera sensor	Image histogram
55	The medial axis provides information about:	Shape structure and symmetry	Image compression	Camera exposure	Pixel storage format
56	Multi-resolution analysis examines an image or shape:	At different levels of detail	At only one resolution	Without pixels	Only in binary form
57	Shape matching attempts to determine:	Similarity between object shapes	Camera temperature	Image file size	Network speed
58	A shape descriptor	Translation,	File renaming	Computer	Network

	should ideally be robust to:	rotation, and scaling		shutdown	disconnection
59	Which method represents an object using its central structural axis?	Medial representation	Contour representation	Histogram representation	Pixel representation
60	Deformable surfaces extend deformable models from:	Curves to surfaces	Images to audio	Pixels to text	Cameras to networks
No .	Question	A	B	C	D
61	The Hough Transform is commonly used for detecting:	Geometric shapes	Audio signals	Database records	Text documents only
62	A standard Hough Transform is particularly effective for detecting:	Lines and other parameterized shapes	Image file formats	Neural network weights	Camera brands

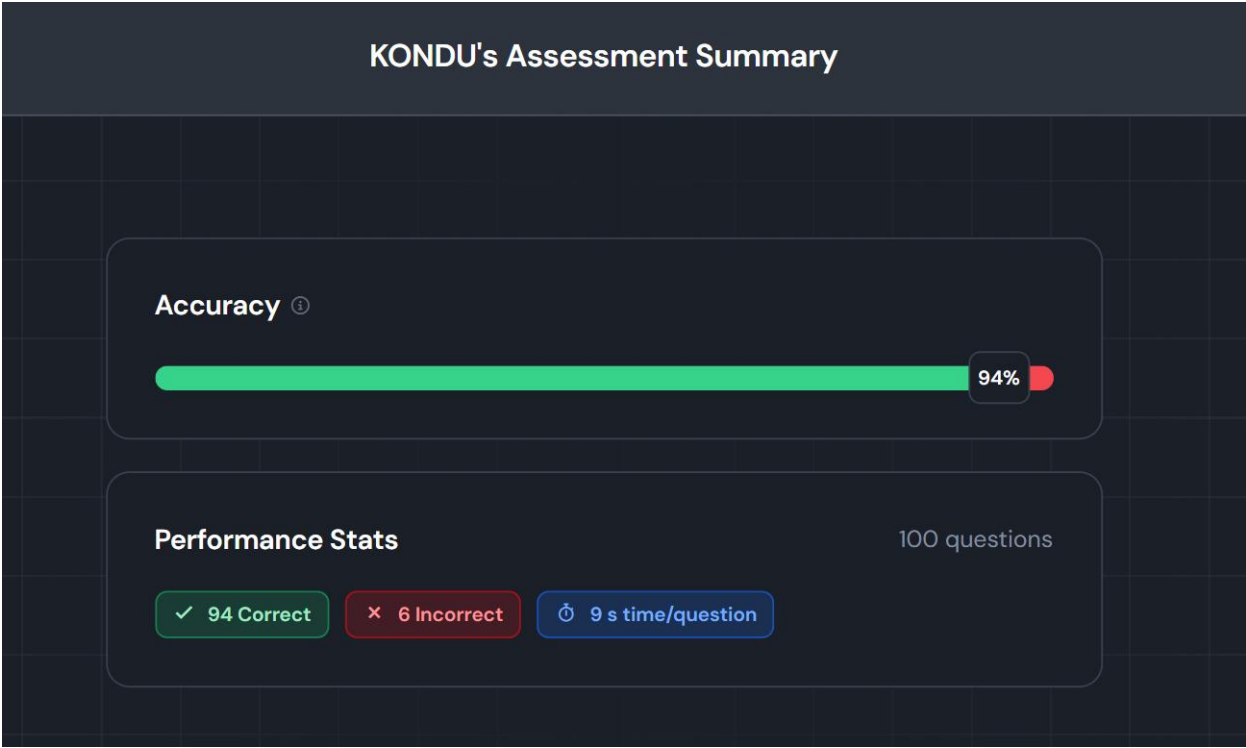
63	Shape correspondence determines:	Relationships between points or parts of shapes	Image compression levels	Camera exposure	Pixel storage format
64	Principal Component Analysis is primarily used for:	Dimensionality reduction	Image acquisition	Camera manufacturing	File compression only
65	PCA transforms data into:	Principal components	Random coordinates	Camera parameters	Pixel labels only
66	Eigenfaces are based primarily on:	PCA	Hough Transform	Optical Flow	GMM
67	Shape priors provide:	Prior knowledge about expected object shapes	Camera calibration data only	Image compression information	Network parameters
68	Deep learning models can learn image features:	Automatically from training data	Only through manual feature selection	Without any data	Only from text files
69	CNN stands for:	Convolutional Neural Network	Computational Network Node	Continuous Neural Normalization	Computer Numerical Network
70	The convolution operation in a CNN is mainly used to:	Extract spatial features	Store image files	Calibrate cameras	Compress videos
71	Pooling layers in CNNs are commonly used to:	Reduce spatial dimensions	Increase image noise	Generate camera models	Convert images into audio
72	Vision Transformers primarily use:	Self-attention mechanisms	Hough voting only	Optical flow only	GMM clustering only
73	GAN stands for:	Generative Adversarial Network	General Analysis Network	Graphical Adaptive Node	Generative Analysis Normalizer
74	A GAN generally consists of:	Generator and discriminator	Encoder and camera	Detector and tracker	Classifier and database
75	Support Vector Machines are commonly used for:	Classification	Image acquisition	Camera calibration	Image resizing
76	The main objective of an SVM is to find:	An optimal separating hyperplane	The brightest pixel	The camera center	The image boundary
77	A Hidden Markov Model is particularly	Sequential data	Image resolution	Camera lenses	Pixel dimensions

	useful for modeling:				
78	Gaussian Mixture Models represent data using:	A combination of Gaussian distributions	A single binary distribution	A single image pixel	A camera matrix
79	The Expectation Maximization algorithm is commonly used to estimate:	Parameters of probabilistic models with latent variables	Camera resolution	Image file size	Object color only

80	Graph-based learning represents data using:	Nodes and edges	Rows and columns only	Pixels and files only	Cameras and lenses
No .	Question	A	B	C	D
81	Photo album organization using computer vision can automatically:	Group and categorize images	Increase camera battery life	Repair computer hardware	Compress operating systems
82	Face detection is the process of:	Locating faces in an image or video	Identifying a person's name directly	Measuring image file size	Calibrating a camera
83	Face recognition attempts to:	Identify or verify a person's identity	Detect image edges only	Remove image noise	Estimate camera motion
84	Eigenfaces are commonly associated with:	Face recognition	Vehicle detection	Stereo matching	Road segmentation
85	Active Appearance Models combine information about:	Shape and appearance	Audio and video	Camera and network	Pixels and file names
86	A 3D face model provides:	Three-dimensional facial shape information	Only image brightness	Only pixel coordinates	Only color histograms
87	Foregroundbackground separation is commonly used in:	Video surveillance	Image printing	File compression	Database indexing
88	Background subtraction is primarily used to:	Separate moving objects from the background	Detect image resolution	Increase camera focal length	Generate image metadata
89	Particle filters are commonly used for:	Tracking objects over time	Image compression	Edge detection only	Camera calibration
90	Chamfer matching is mainly used for:	Shape matching	Image enhancement	Video compression	Audio recognition
91	Occlusion occurs when:	One object partially or completely blocks another	Image brightness increases	A camera changes resolution	Pixels are converted to grayscale
92	Multi-camera systems can provide:	Multiple viewpoints of a scene	Only onedimensional information	Audio-only information	Image compression only
93	Human gait analysis studies:	Patterns of human walking	Facial color	Road surface texture only	Camera lens properties
94	An in-vehicle vision system can be used for:	Road and pedestrian detection	Database administration	Audio compression	Operating system management
95	Roadway localization in autonomous vehicles helps determine:	The drivable road area	The vehicle's software version	The camera brand	The image file type
96	Road marking detection is important	Lane and roadway understanding	Face recognition	Medical diagnosis	Photo album sorting
	for:				

97	Road sign recognition is an important component of:	Autonomous driving systems	Image compression systems	Database systems	Audio processing systems
98	Medical image segmentation is used to:	Separate anatomical structures or regions of interest	Increase internet speed	Detect road markings	Organize photo albums
99	Augmented Reality overlays:	Digital information onto the real-world environment	Audio signals onto databases	Medical records onto cameras	Image files onto operating systems
100	Mixed Reality combines:	Physical and digital environments with interactive elements	Only twodimensional images	Audio and text files only	Camera calibration and image compression

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor	Some correct answers; frequent	Mostly incorrect answers



Total Marks: 94/100

